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$$f(-x) = -\sin x = -f(x) \Rightarrow f(x) \text{ is oneven} \Rightarrow a_0 = a_n = 0$$

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^\pi \sin x \sin nx dx \\ &= \frac{1}{\pi} \int_0^\pi (\cos(x-nx) - \cos(x+nx)) dx \\ &= \frac{1}{\pi} \left[\frac{\sin(x-nx)}{1-n} - \frac{\sin(x+nx)}{1+n} \right]_0^\pi \\ &= \frac{1}{\pi} \left(\frac{\sin(\pi-n\pi)}{1-n} - \frac{\sin(\pi+n\pi)}{1+n} \right) \\ &= \frac{1}{\pi} \left(\frac{0}{1-n} - \frac{0}{1+n} \right) = 0 \Rightarrow b_n = 0 \text{ voor } n \neq 1 \\ b_1 &= \frac{2}{\pi} \int_0^\pi \sin^2 x dx \\ &= \frac{1}{\pi} \int_0^\pi (1 - \cos 2x) dx \\ &= \frac{1}{\pi} \left[x - \frac{\sin 2x}{2} \right]_0^\pi = 1 \\ &\Rightarrow \sum_{n=1}^{\infty} b_n \sin nx \text{ met } b_n = 0, b_1 = 1 \\ &\Rightarrow \text{fourierreeks: } \sin x \end{aligned}$$

References